Lecture 21
Summary and Outlook

Algorithms, AI & Data Science II

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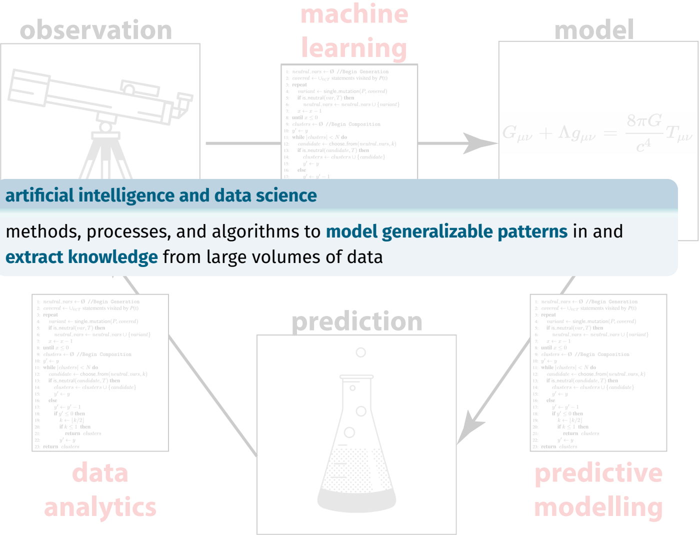
Notes:

- **Lecture L21: Summary and Outlook** 21.07.2025
- **Educational objective:** We summarize the course, give an outlook to advanced topics that you can study in follow-up courses and textbooks, and discuss the results of the lecture evaluation.
- **Handout version:** *2025/06/24 15:27:41*

Motivation



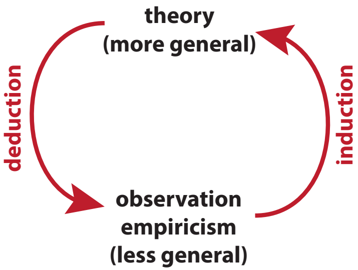
artificial intelligence and data science

methods, processes, and algorithms to **model generalizable patterns** in and **extract knowledge** from large volumes of data

Notes:

Inductive vs. deductive reasoning

- extending the simplistic view given in the motivation, we considered both **inductive and deductive methods of knowledge extraction**
- **deductive approaches** include symbolic AI like, e.g., logical inference
- **inductive approaches** include statistical inference and machine learning
- inductive approaches to AI have recently become much more common (and successful)
- <u>but:</u> **both approaches have their justification in AI**



Notes:

Algorithms, AI, and Data Science II

Advanced Topics in AI and Data Science

this course

- Logics, Reasoning, Computation
- Probabilistic Reasoning and Inference
- Information Compression Pattern Recognition
- Graphs Topology Geometry

Algorithms, AI & Data Science I

Data Structures
Algorithms and Complexity
Search and Optimization

Notes:

Final syllabus

Part I - Logics, Reasoning, and Computation

- L01 - Motivation → 28.04.2025
- L02 - Formal Foundations of Computing → 30.04.2025
- L03 - Word Problems and Automata → 05.05.2025
- L04 - Turing Machines and Computability → 06.05.2025
- L05 - Propositional Logic and Boolean Algebra → 14.05.2025
- L06 - Satisfiability in Propositional Logicv → 19.05.2025
- L07 - Logical Inference → 21.05.2025
- L08 - First-Order Logic → 26.05.2025

Sir Arthur Conan Doyle
1859 – 1930

Part II - From Logical to Probabilistic Reasoning

- L09 - Probabilistic Foundations → 28.05.2025
- L10 - Statistical Modelling and Inference → 11.06.2025
- L11 - Bayesian Inference → 16.06.2025
- L12 - Hypothesis Testing → 18.06.2025
- L13 - Challenges in Data Science → 23.06.2025

logical inference

"He was, I take it, the most perfect reasoning and observing machine that the world has seen [. . .]."
→ Sherlock Holmes, A Scandal in Bohemia

image credit: Wikimedia Commons, public domain

Notes:

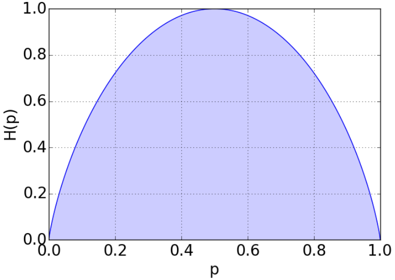
Final syllabus

Part III - Information, Compression, Pattern Recognition

- L14 - Markov Chains and Random Walks → 25.06.2025
- L15 - Markov Chain Convergence and PageRank → 30.06.2025
- L16 - Markov Chain Monte Carlo → 02.07.2025
- L17 - Information Theory and Data Compression → 07.07.2025
- L18 - Description Length Minimization → 09.07.2025

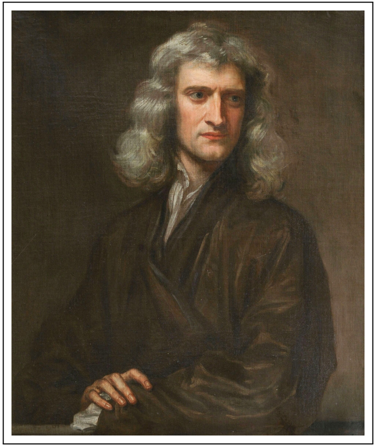
Part IV - Graphs and Topology

- L19 - Graphs, Laplacians and Minimum Cuts → 14.07.2025
- L20 - Flow Maximinization in Graphs → 16.07.2025
- L21 - Summary and Conclusion → 21.07.2025



Notes:

Is that all there is to know?



Isaac Newton

> "What we know is a drop. What we do not know is an ocean."
> Isaac Newton

image credit: Portrait of Isaac Newton by Godfrey Kneller, public domain

Notes:

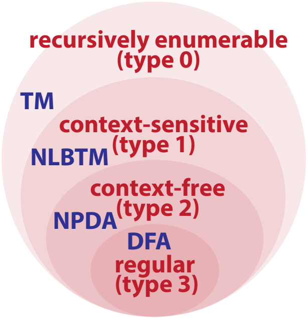
Outlook: Formal languages and NLP

- in L02 and L03 we introduced **formal languages and formal grammars**
- **Chomsky hierarchy** is foundation for **computational linguistics and natural language processing**
- we studied which automata models are needed to solve **word decision problems** for different formal languages

further topics

- regular expressions and **Kleene's theorem**
- **pumping lemma** for regular and context-free languages
- **Backus-Naur** form for context-free grammars
- **CYK algorithm** for context-free languages in Backus-Naur form

- see course **Theoretische Informatik** (Prof. Glasser)



Chomsky hierarchy with automata models that solve word decision problem → L03

Notes:

Outlook: Automata, computability, and complexity

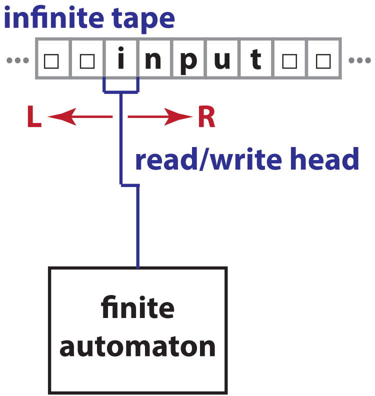
- in L03 we explored different **automata models** to solve word decision problems

- in L04 we introduced (universal) **Turing machines** as basis for our notion of **computability**

further topics

- **LOOP, WHILE, GOTO-computability**
- **primitive recursive** and μ-recursive functions
- **Ackermann function**
- **Post's correspondence problem**
- **Gödels incompleteness theorem**
- P, NP, **NP-completeness**, and reducibility

- see courses **Theoretische Informatik** and **Berechenbarkeitstheorie** (Prof. Glasser)



Turing machine → L03

Notes:

Outlook: From first- to higher-order logic

- In L05 we introduced **propositional and first-order logic**
- in L06 and L08 we studied the **satisfiability problem** in propositional and first-order logic
- in L07 and L08 we explored **logical inference** in propositional and first-order logic

further topics

- **unification algorithm** for first-order inference
- **backward chaining** in first-order logic
- **higher-order and modal logic**
- **knowledge representation** and **ontologies**
- **logic programming** and **PROLOG**

- see course **Logik für Informatiker** (Prof. Seipel)

$$(X_0 \vee X_1 \vee X_2) \wedge (\neg X_1 \vee \neg X_2) \wedge X_0 \wedge \neg X_0$$

resolution algorithm

procedure RESOLUTION(F)
  choose X that also appears as $\neg X$ in F
  for clauses c_i with $X \in c_i$ **do**
    for clauses c_j with $\neg X \in c_j$ **do**
      $r \leftarrow c_i - \{X\} \vee c_j - \{\neg X\}$
      $F \leftarrow$ add_clause(F, r)
    end for
  end for
  for clauses c_i with $X \in c_i$ or $\neg X \in c_i$ **do**
    $F \leftarrow$ remove_clause(F, c_i)
  end for
end procedure

$\longrightarrow$ L06

Notes:

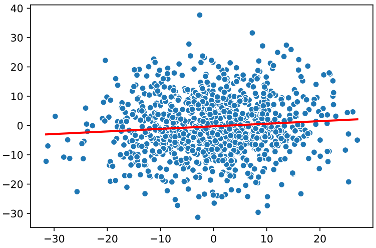
Outlook: Pattern recognition and dimensionality reduction

- in L09 – L11 we introduced foundations of **statistical modelling** as well as **frequentist and Bayesian inference**
- in L11 we took a statistical inference perspective on **linear regression**
- in L12 – L13 we explored **hypothesis testing in data**

further topics

- **MCMC sampling** in Bayesian inference
- **expectation maximization** for inference wirth hidden variables
- cluster detection with **gaussian mixture model** and **DBSCAN**
- classification with **logistic regression** and **neural networks**
- dimensionality reduction with **principal component analysis** (PCA), **singular value decomposition** (SVD), or **t-SNE**

- see courses **Introduction to Artificial Intelligence** (Prof. d'Eramo) and **Data Mining** (Prof. Hotho)



linear regression → L11

Notes:

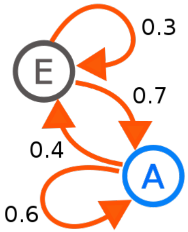
Outlook: Probabilistic graphical models

- in L14 and L15 we introduced **Markov models** as model for **stochastic processes**
- in L16 we considered **algorithmic applications** of Markov chains
- in L13 we considered **causal diagrams** where directed edges capture causal relationships

further topics

- **higher-order** Markov models
- **hidden Markov models** (HMM)
- **inference** for Markov chains
- **Bayesian networks** and **causal inference**
- temporal models: **Kalman filters** and **dynamic Bayesian networks**

- see courses **Data Mining** and **Time Series and Anomaly Detection** (Prof. Hotho)



graph representation of Markov chain with two states A and E

→ L13

image credit: Wikimedia Commons, User Joxemai4, CC-BY-SA

Notes:

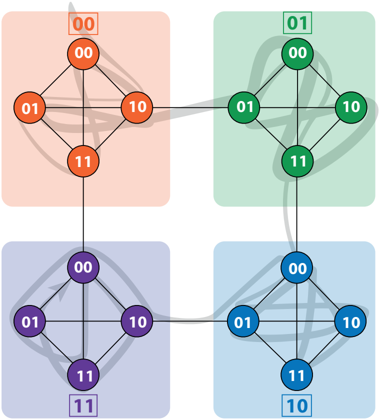
Outlook: Graph algorithms and graph learning

- in L18 we explored **cluster detection in graphs**
- in L19 studied graph Laplacians and **minimal cuts**
- in L20 we explored the Ford-Fulkerson method for **flow maximization** and proved the **max-flow min-cut theorem**

further topics

- **maximum matchings** in (bipartite) graphs
- **graph coloring** and labeling problems
- **graph drawing** algorithms
- **minimum spanning trees**
- **traveling salesman** problem (NP-complete)

- see course **Algorithmische Graphentheorie** (Prof. Wolff), **Statistical Network Analysis** and **Machine Learning for Complex Networks** (Prof. Scholtes)



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flow compression with InfoMap → L17

Notes:

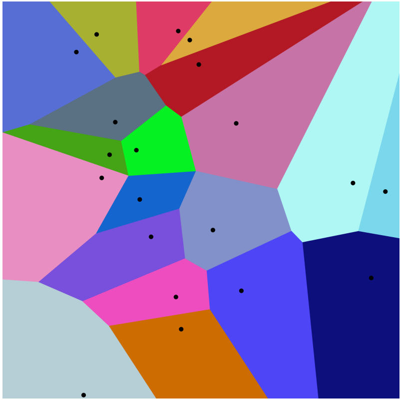
Outlook: Algorithmic geometry

- in our course, we did not consider specific **algorithms for geometric problems**
- algorithms for geometric problems are important in **computer graphics, robotics, and point cloud processing**

further topics

- **Voronoi diagrams**
- **Delaunay triangulation**
- **convex hull computation**
- **closest pair calculation**

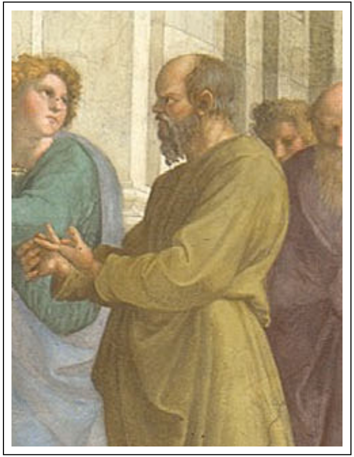
- see course **Algorithmische Geometrie** (Dr. Klemz)



Voronoi diagram of a point cloud

Notes:

Kindling the flame …



Sokrates

image credit: Excerpt from Scuolo di Athena, public domain

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> "Education is the **kindling of a flame**, not the filling of a vessel."
> Sokrates

Notes:

Advanced courses at JMU

Selected lectures

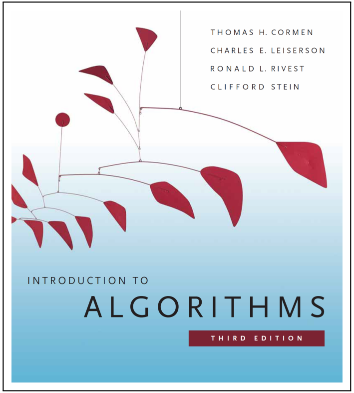
- Lecture: **Computer Vision**, Prof. Dr. Timofte
- Lecture: **Machine Learning for NLP**, Prof. Dr. Hotho
- Lecture: **Natural Language Processing**, Prof. Dr. Glavas, Prof. Dr. Hotho
- Lecture: **Theory of Machine Learning**, N.N.
- Lecture: **Information Retrieval**, Prof. Dr. Hotho
- Lecture: **Time Series and Anomaly Detection**, Prof. Dr. Hotho
- Lecture: **Machine Learning for Complex Networks**, Prof. Dr. Scholtes
- Lecture: **Statistical Network Analysis**, Prof. Dr. Scholtes
- Lecture: **Cognitive Systems**, Prof. Dr. N.N.
- Lecture: **Knowledge-based Systems**, Prof. Dr. Puppe
- Lecture: **Image Processing and Computational Photography**, Prof. Dr. Timofte
- Lecture: **3D Point Cloud Processing**, Prof. Dr. Nüchter
- Lecture: **Computational Complexity**, Prof. Dr. Wolff
- Lecture: **Algorithmic Graph Theory**, Prof. Dr. Wolff
- Lecture: **Algorithmic Geometry**, Dr. Klemz
- Lecture: **Optimization for Machine Learning**, N.N.

seminars

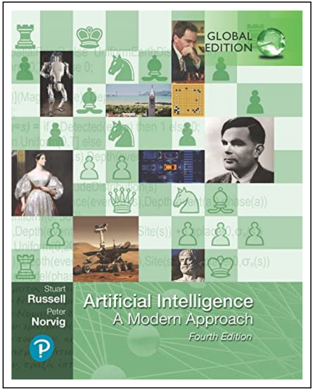
- Seminar: **Ausgewählte Kapitel des Machine Learning**, Prof. Hotho
- Seminar: **Aktuelle Trends in der Künstlichen Intelligenz**, Prof. Puppe
- Seminar: **Computer Vision**, Prof. Timofte
- Seminar: **Machine Learning for Complex Networks**, Prof. Dr. Scholtes
- Seminar: **Data, AI, and Society**, Prof. Dr. Scholtes
- Seminar: **Vision and Language**, Prof. Dr. Glavas and Prof. Dr. Timofte

Notes:

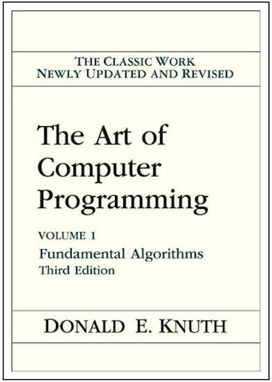
Further reading ...



TH Cormen, CE Leiserson, RL Rivest, C Stein: **Introduction to Algorithms**, 4th edition, 2022



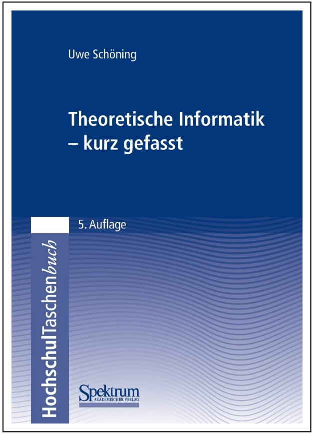
S Russel, P Norvig: **Artificial Intelligence: A Modern Approach**, 4th edition, 2020



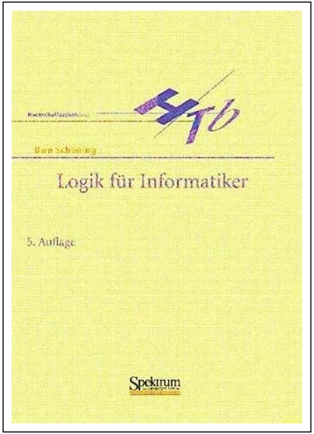
DE Knuth: **The Art of Computer Programming**, Vol. I – IV, 1968 – ?

Notes:

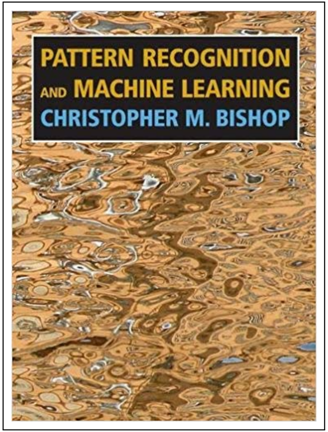
Further reading ...



U Schöning: **Theoretische Informatik - kurz gefasst**, 5th edition, 2008



U Schöning: **Logik für Informatiker**, 5th edition, 2000



CM Bishop: **Pattern Recognition and Machine Learning**, 1st edition, 2006

Notes:

Information on final exam

Q & A Session **today!**

We answer questions that you identified during the exam preparation and offer to repeat selected parts of the lectures.

Final Exam **23.07.2025**

- **Wednesday, July 23rd 2025**, from **14:30 - 16:30**, **Turing lecture hall**
- written, **closed-book exam** of 120 minutes
- no specific `python` questions (only "pseudocode" level)
- approx. 30 mins worth of **multiple-choice questions**
- approx. 90 minutes worth of open answer questions
- you can bring a **hand-written double-sided A4 "cheat" sheet**
- please **bring a copy of your exam registration** (registration number needed for exam cover sheet)

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Notes:

Thank you!

- we covered basic foundations of **algorithms, AI, and data science**
- we **implemented and explored** them in `python`

some statistics

- **21 lectures** with **540 slides** generated by **33,255 lines of LaTeX code**
- **7200 lines of python code** producing **734 figures and illustrations**
- **32 group exercises**
- practice sessions with **42 jupyter notebooks**

- a **complete script is available** on WueCampus
- **thank you** for your attention and active participation!
- we hope to see many of you in **further courses, seminars, or your BSc thesis**!

Notes: